

Heart Disease AI System

PRCP-1016 — Prediction Report

Domain: Healthcare | Type: Binary Classification

Dataset: 180 patients | Features: 13 clinical variables

Best Model: Random Forest | ROC-AUC: 0.95

1. Dataset Summary

Total patients: 180

No Heart Disease (0): 100 (55.6%)

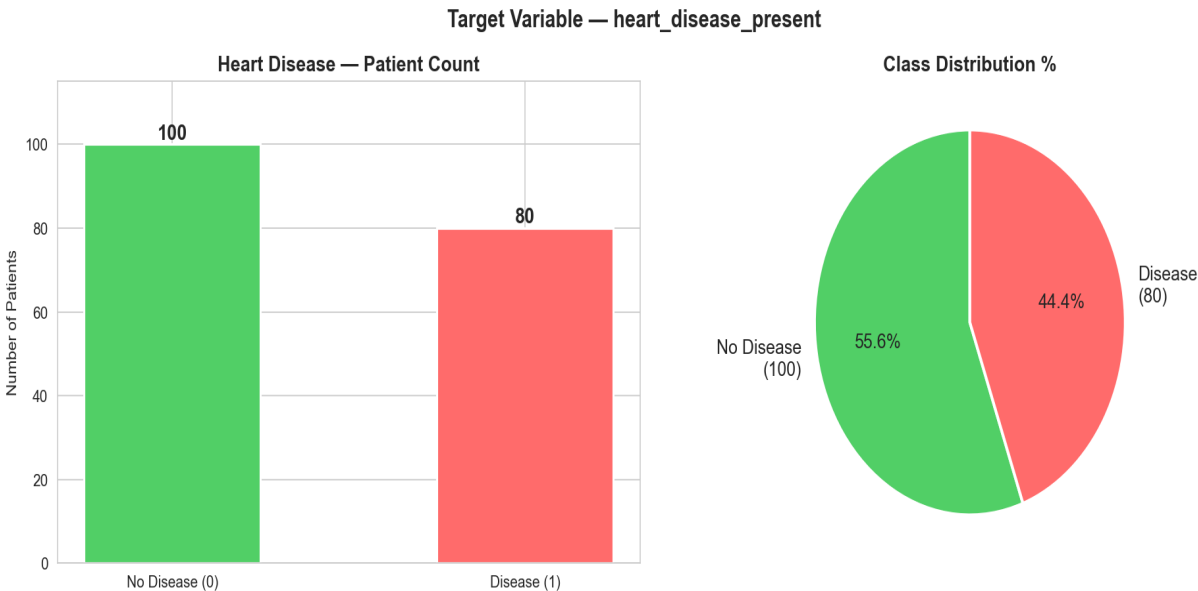
Heart Disease (1): 80 (44.4%)

Missing values: None

Only categorical column: thal (normal / fixed_defect / reversible_defect)

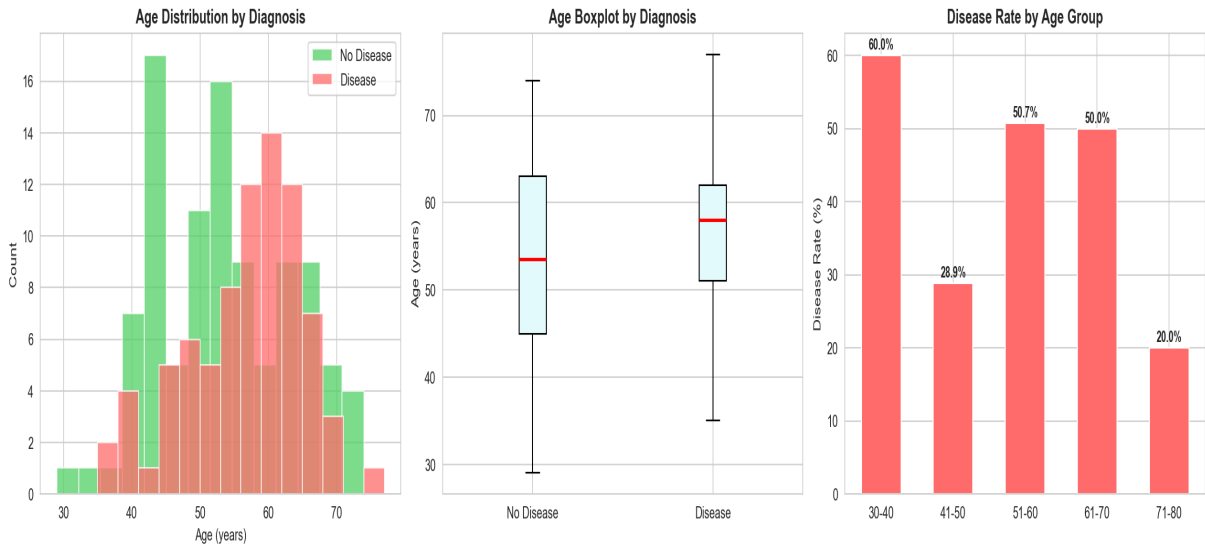
2. EDA Charts

Target Variable Distribution

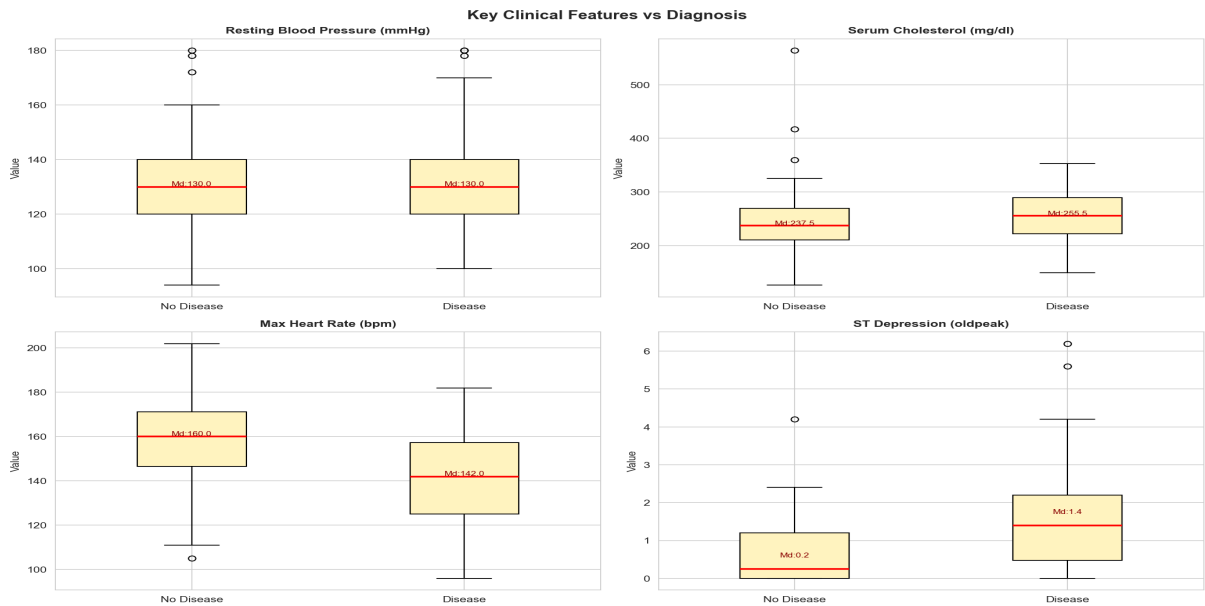


Age Analysis

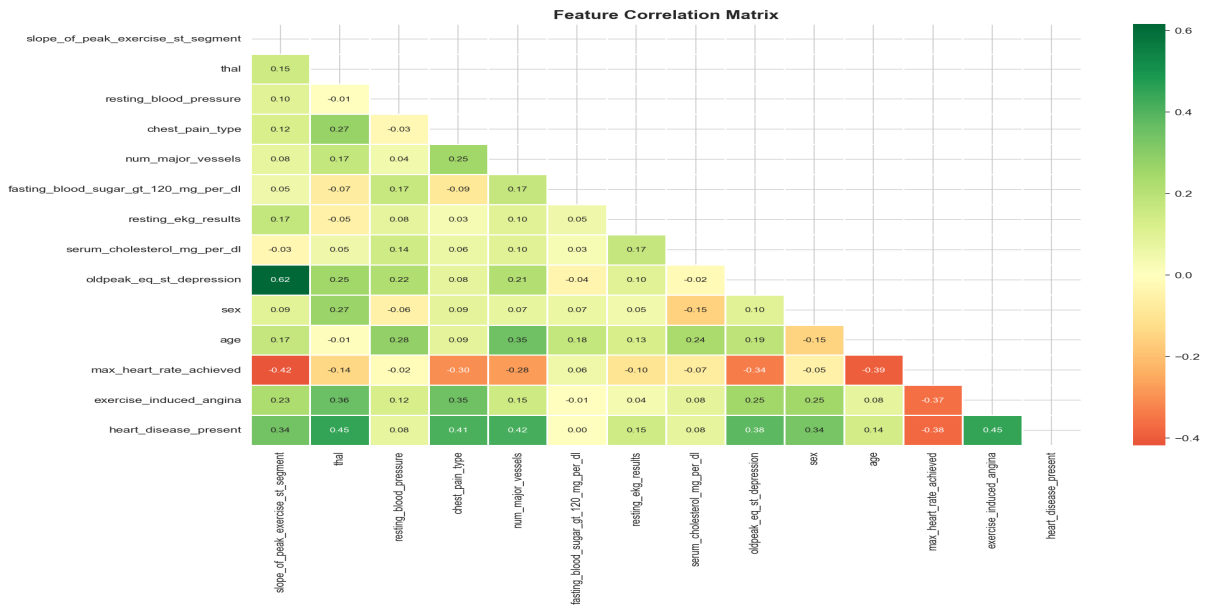
Age vs Heart Disease



Clinical Features vs Diagnosis



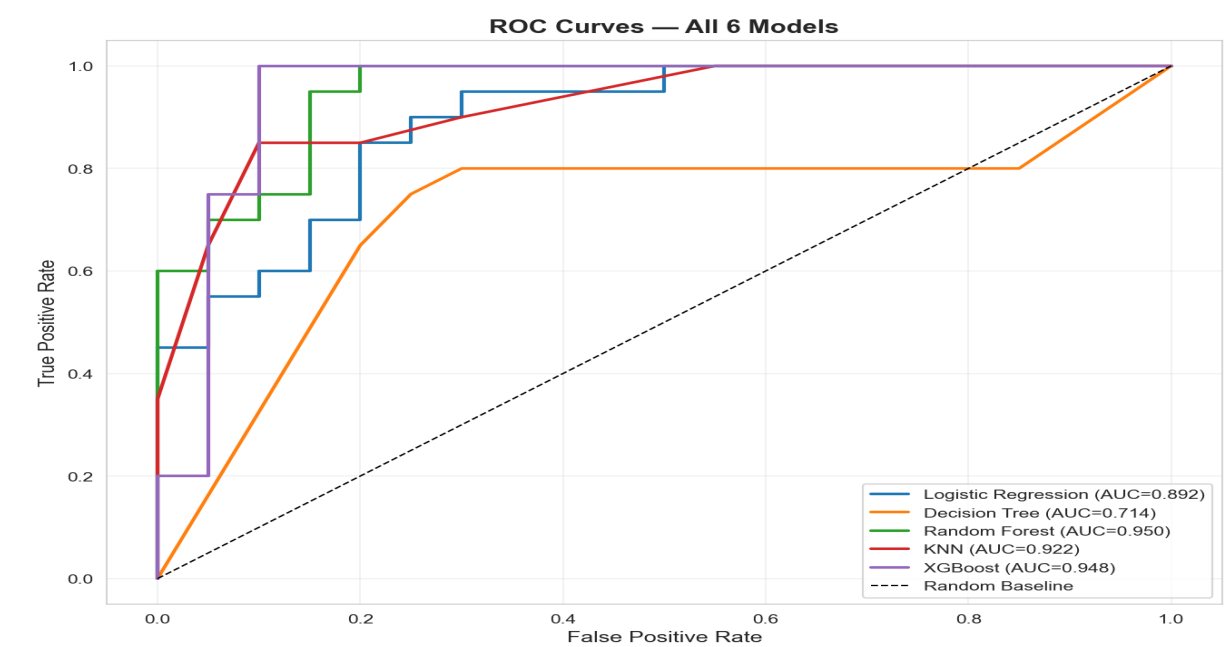
Feature Correlation Heatmap



3. Model Comparison Report

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.85	0.85	0.85	0.85	0.95
XGBoost	0.925	0.8696	1.0	0.9302	0.9475
KNN	0.825	0.8095	0.85	0.8293	0.9225
Logistic Regression	0.825	0.8095	0.85	0.8293	0.8925
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Decision Tree	0.75	0.7273	0.8	0.7619	0.7138
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ROC Curves — All Models



Confusion Matrices — All Models

4. Challenges & Solutions

Challenge 1 — Small Dataset (180 rows)

Used stratified train-test split and 5-fold cross-validation to maximise use of limited data.

Challenge 2 — Class Imbalance (55.6% vs 44.4%)

Applied SMOTE oversampling and `class_weight=balanced` to prevent bias toward the majority class.

Challenge 3 — Categorical Column (thal)

LabelEncoder applied. Tree models handle ordinally-encoded categoricals natively.

Challenge 4 — Feature Scaling

StandardScaler applied only for Logistic Regression, SVM, and KNN. Tree-based models do not require scaling.

Challenge 5 — Metric Selection

Prioritised Recall and ROC-AUC over raw accuracy. In medical diagnosis, False Negatives (missed disease) are more dangerous than False Positives.

5. Hospital Recommendations

Critical Indicators (Act Immediately)

- Reversible defect on thallium stress test: immediate cardiology referral.
- ST depression (oldpeak) > 1.5 mm: urgent cardiac workup.
- Asymptomatic chest pain (Type 4): highest disease rate — do not dismiss silent presentations.

Proactive Screening

- Patients with 2+ major vessels colored by fluoroscopy: place on cardiac monitoring.
- Males over 50: mandatory annual ECG.
- Low max heart rate during stress test: reduced cardiac reserve — investigate.

Deployment Suggestion

Deploy Random Forest (ROC-AUC=0.95) as a triage risk-scoring tool. Patients with predicted probability > 70% are fast-tracked to cardiology. Retrain quarterly as new data arrives.